

Lannawill Caruth

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EDUCATION

Vanderbilt University | Bachelor of Science, Engineering Science, May 2023

Minors: Scientific Computing | Medicine, Health, & Society

EXPERIENCE

- **Junior Statistician | University of Pennsylvania Perelman School of Medicine - Setia-Verma Lab** August 2023 - Present
 - Developed bioinformatics analysis pipelines using Python and Nextflow. Managed GitHub repository for public distribution of pipelines
 - Assisted senior staff and graduate students in project management and development for integrating multi-omics and EHR data for risk prediction, primarily for women's health conditions
 - Performed association testing (PheWAS, GWAS, ExWAS) for clinical collaborators investigating genomic connections to phenotypes of interest
 - Presented results from ongoing research efforts at national and local conferences.
- **Undergraduate Research Assistant | Vanderbilt University Medical Center - Barnado Lab.** January 2022 – May 2023
 - Collaborated on a data analysis effort to determine the connection between a positive antibody test and autoimmune disease
 - Utilized R and Python to obtain summary statistics from multiple data sets
 - Created training materials to provide introductory data analysis training to research assistants

PUBLICATIONS

Journal Articles

- Furquim, C. P., **Caruth, L.**, ..., Verma, S. S., & Teles, F. (2025). Developing predictive models for periodontitis progression using artificial intelligence: A longitudinal cohort study. *Journal of clinical periodontology*, 52(10), 1478-1490.
- Barnado, A., ..., **Caruth, L.**, ..., et al. (2023). Identifying antinuclear antibody positive individuals at risk for systemic autoimmune disease: Development and validation of a real-time risk model. *Arthritis & Rheumatology*, 75, 236–239.

- Barnado, A., ..., **Caruth, L.**, ..., et al. (2024). Identifying antinuclear antibody-positive individuals at risk for developing systemic autoimmune disease: Development and validation of a real-time risk model. *Frontiers in Immunology*, 15. <https://doi.org/10.3389/fimmu.2024.1384229>
- Verma, S. S., **Caruth, L.**, ..., et al. (2024). Development of a comprehensive predictive model for primary open-angle glaucoma in individuals of African ancestry. *Investigative Ophthalmology & Visual Science*, 65(7), 2762–2762.
- Guare, L. A., Das, J., **Caruth, L.**, & Setia-Verma, S. (2024). Social determinants of health and lifestyle risk factors modulate genetic susceptibility for women’s health outcomes. In *Biocomputing 2025: Proceedings of the Pacific Symposium* (pp. 296–313).

Preprints

- Deutsch, C. J., Sykes, A. E., **Caruth, L.**, Setia-Verma, S., & Hoshi, T. (2025). Disease-associated Kv1.3 variants are energy-compromised with impaired nascent chain folding. *bioRxiv*.
- Guare, L. A., Das, J., **Caruth, L.**, Rajagopalan, A., Akerele, A. T., Brumpton, B. M., ... & Setia-Verma, S. (2024). Expanding the genetic landscape of endometriosis: Integrative-omics analyses specify key pathways from a multi-ancestry study of over 900,000 women. *medRxiv*.

Selected Conference and Invited Presentations

- “Evaluating the effectiveness of machine learning models using polygenic risk scores and clinical factors for breast cancer risk stratification”, American Association of Cancer Research, Chicago, IL, April 2025.
- “Evaluating effectiveness of polygenic risk scores in African ancestry breast cancer patients from Penn Medicine BioBank”, American Society of Human Genetics, Denver, CO, November 2024

LEADERSHIP, PROFESSIONAL DEVELOPMENT, & AFFILIATIONS

Society of Women Engineers | August 2019 – May 2023

Collegiate Chapter Vice President

- Organized monthly General Body Meetings focused on networking and professional development for women engineers.
- Represented the chapter at the SWE National Conference.

Multicultural Leadership Council | August 2021 – May 2023

- Managed a \$15K event budget to plan formal events serving Vanderbilt’s multicultural community.
- Chaired the underclassmen mentorship group, training students on scheduling and planning meetings with administration, organizing events, and navigating internal communication and expense systems.